Investigating Appointment Trends, Regional Variation, and Service Efficiency in NHS Healthcare Delivery

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Background and context of the business

Healthcare systems face fluctuating appointment demand across regions and time.

Understanding these variations is critical for efficient service delivery.

Data-driven insights can support better planning and reduce operational pressure.

Objectives

Identify seasonal trends
Analyse patterns in appointment volumes over time to detect peak and low-demand periods.

Assess regional demand
Compare appointment volumes across locations to highlight disparities and resource needs.

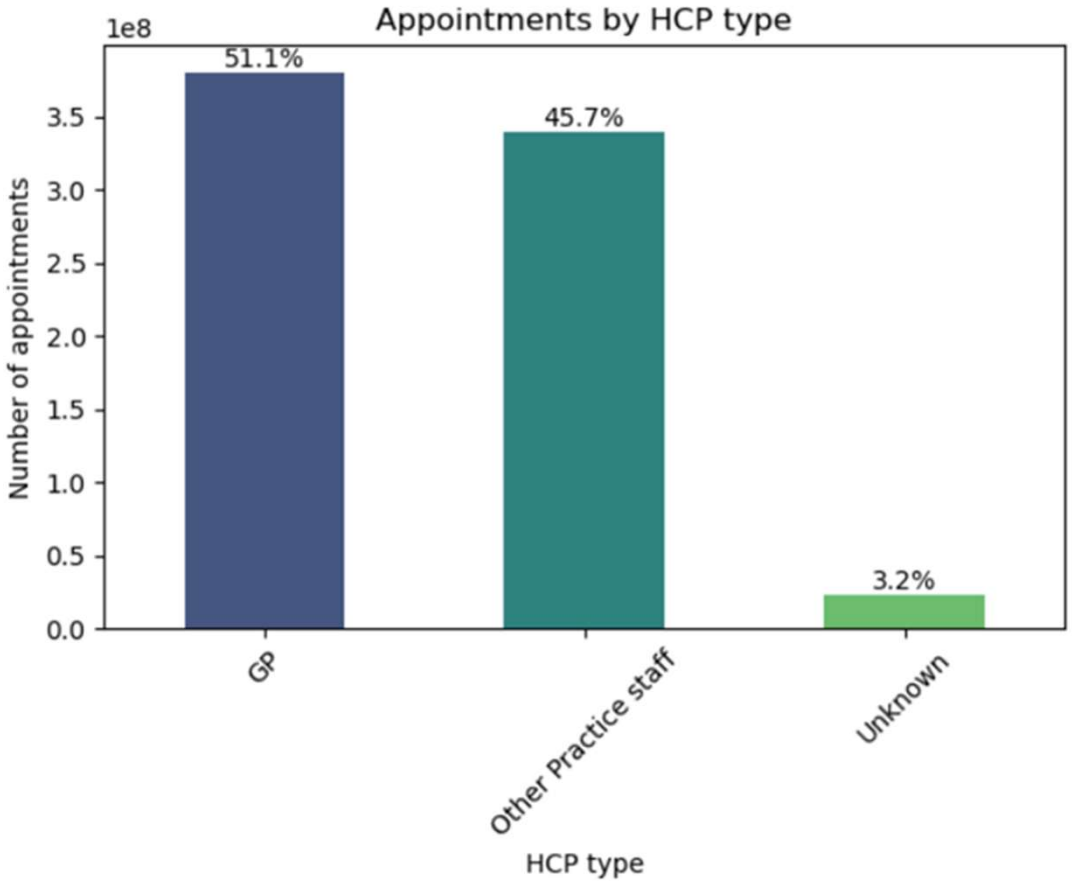
Improve service planning
Use appointment data to optimise staffing, allocate resources effectively, and reduce system strain.

Workforce Structure

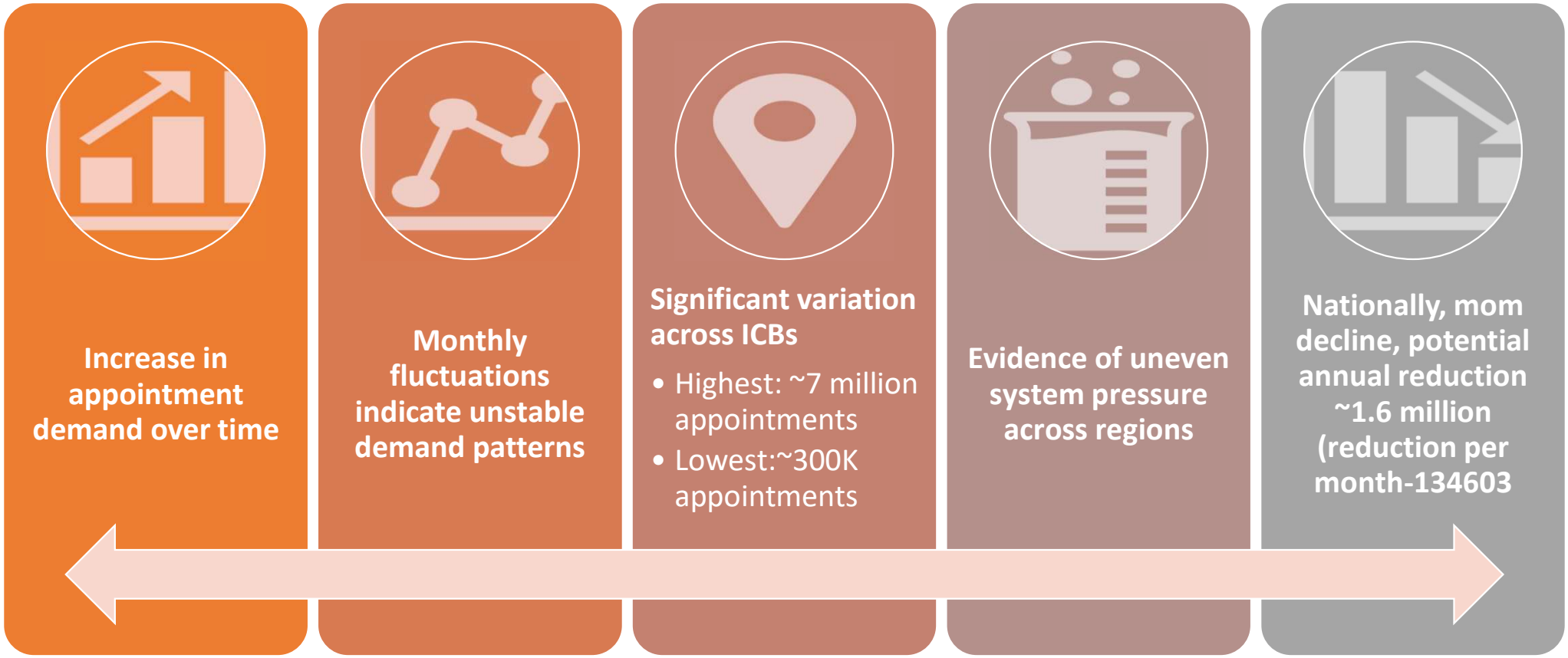
GP and Other Practice Staff deliver **~97% of all appointments**

GP-led system with **growing support from wider clinical staff**

Minimal “Unknown” category (~3%): **minor data gaps**

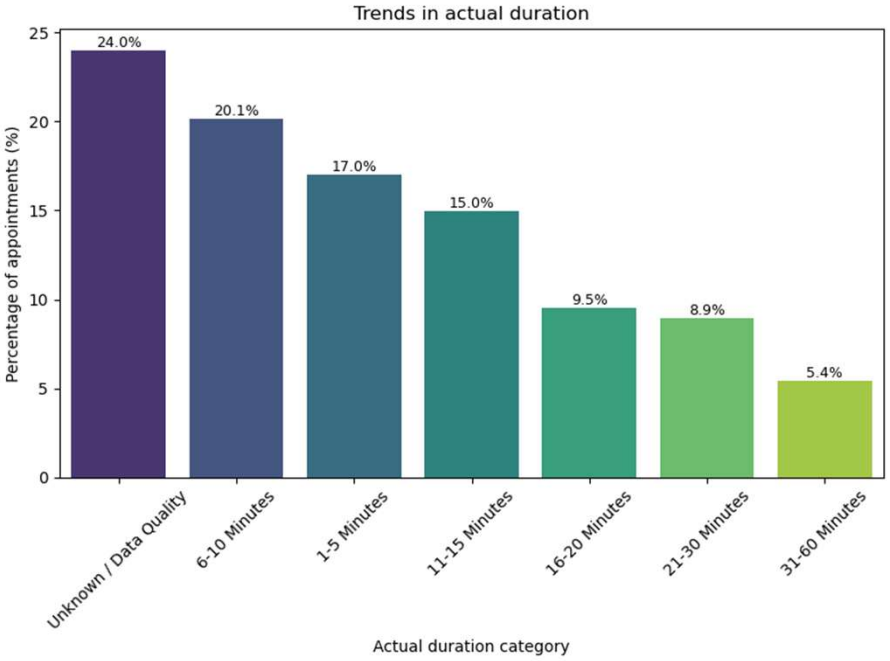


Growth Demand & Pressure



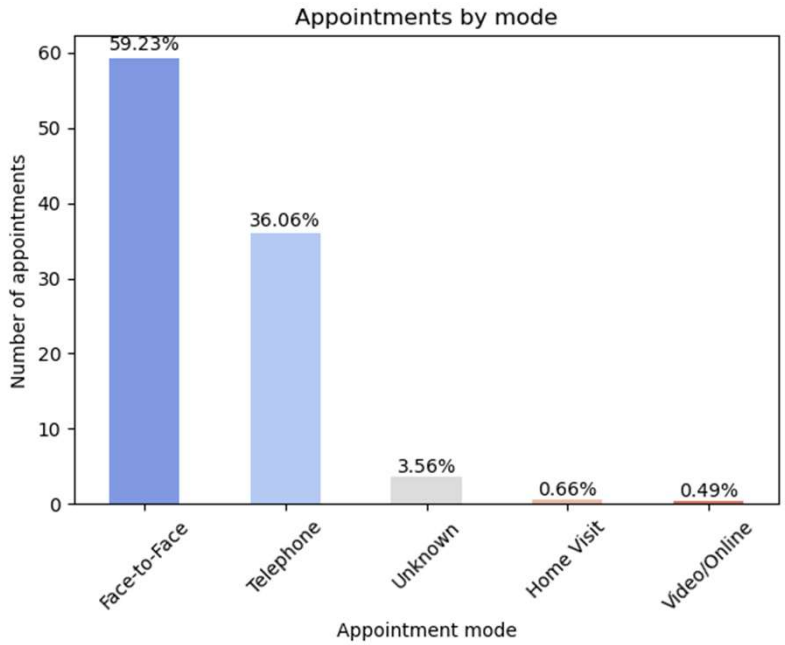
Service Delivery Patterns

Majority of appointments are short <15 minutes

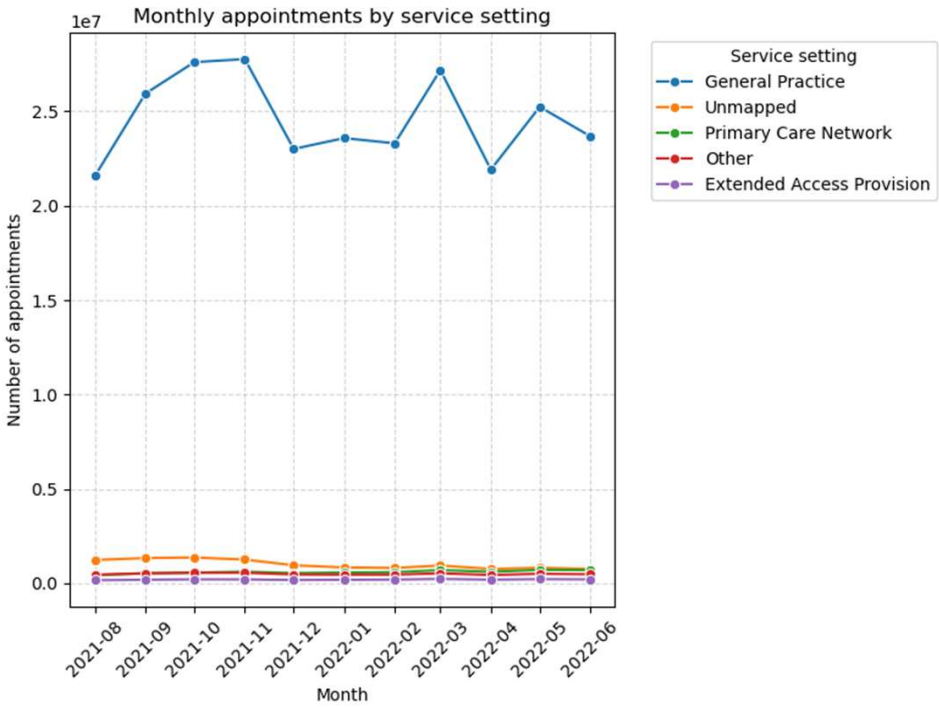


Face-to-face 59% and telephone 36%

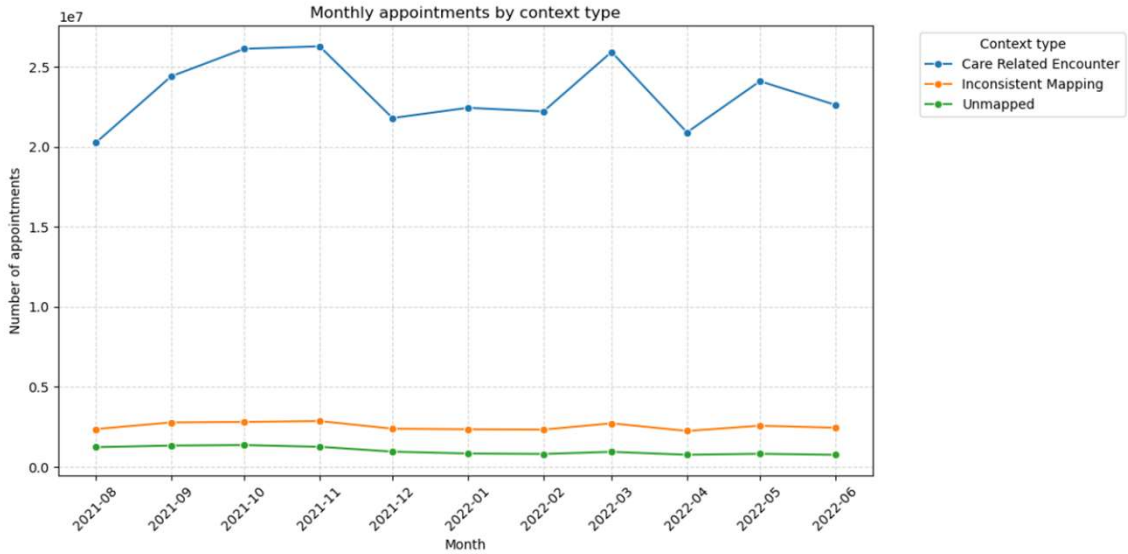
Very low use of video/online < 1%



General Practice & routine consultations are the main service type



System focused on high-volume, routine care delivery

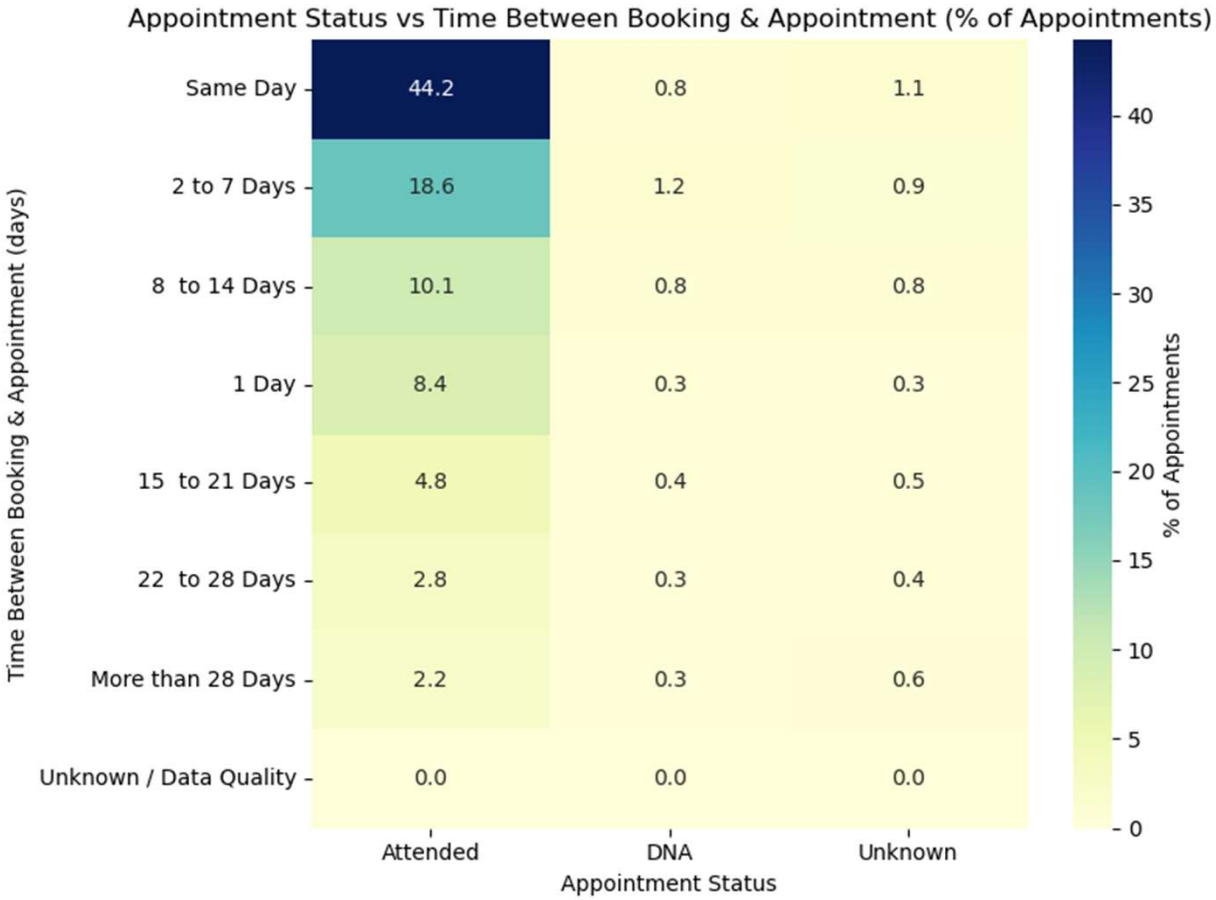


System Performance

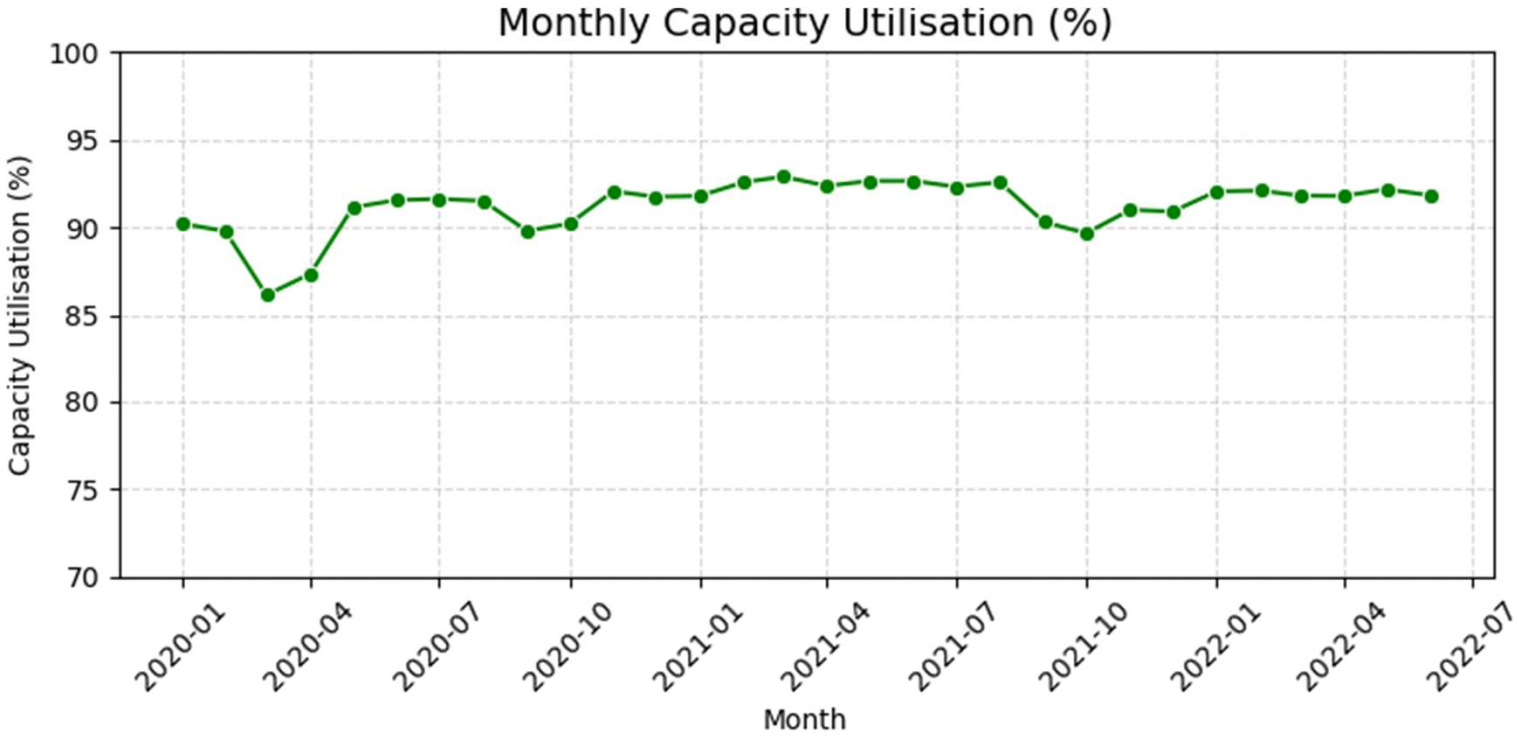
Most appointments occur **same day or within 7 days**

High attendance rates, very low DNA

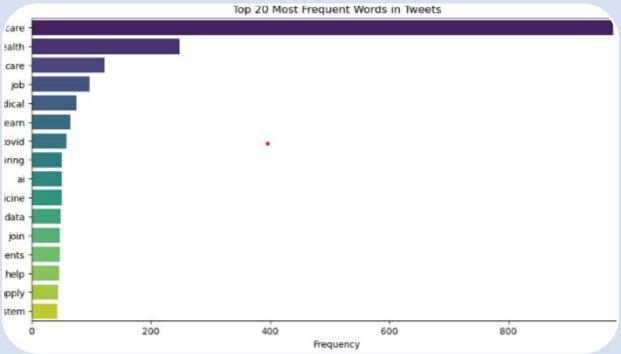
🔥 **Financial impact: ~£50.4 million lost due to missed appointments**

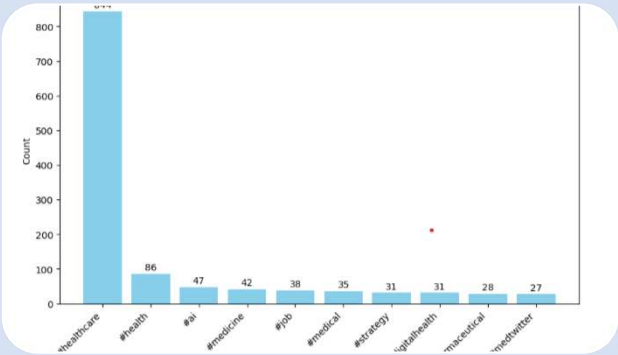


Capacity utilisation ~91%
Indicate that system near full capacity



Tweet Analysis Overview





Sentiment Category	Count
Positive	618
Neutral	449
Negative	107

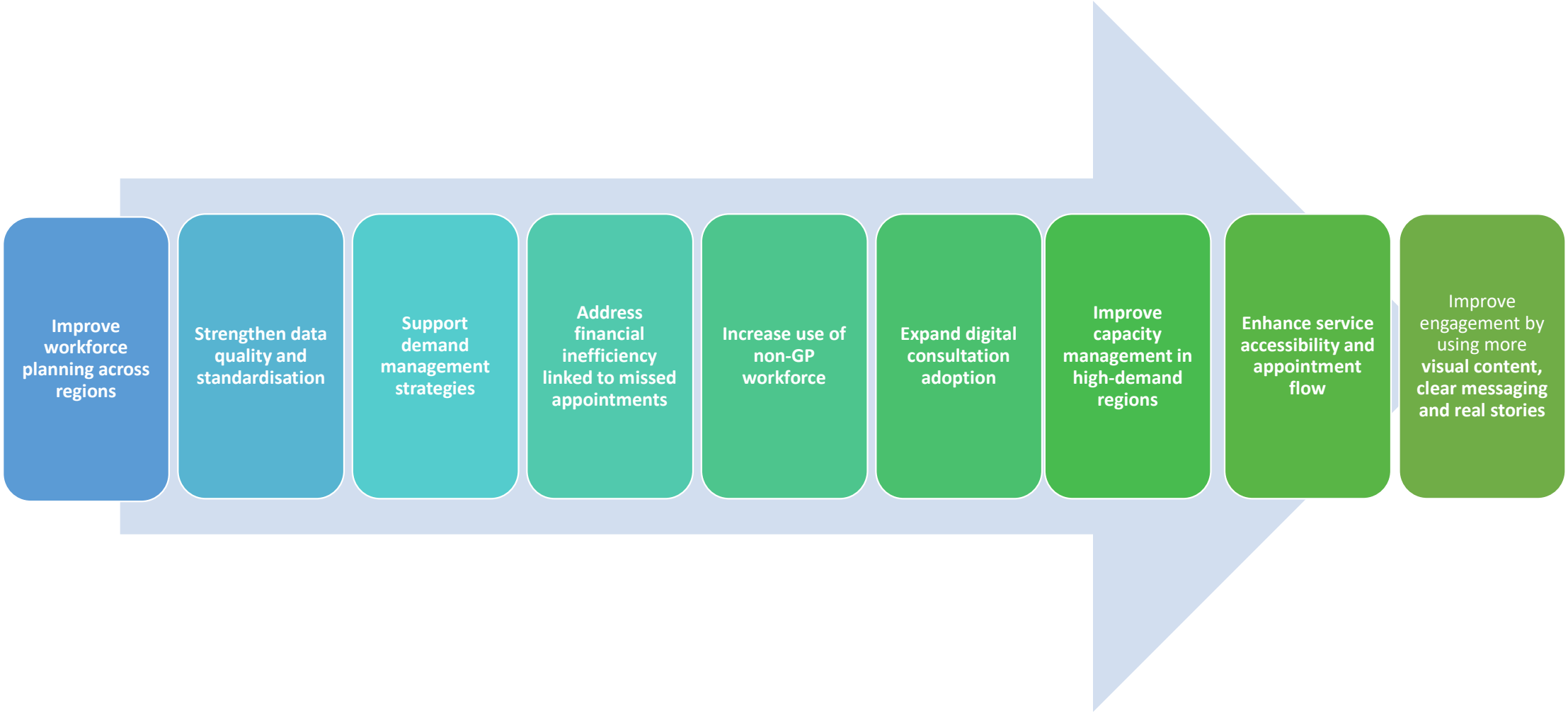
Tweets are mainly focused on **healthcare topics, jobs, AI and medical related**.

Engagement is **highly uneven,** most tweets receive low interaction, while a few gain very high visibility.

Sentiment is mostly **positive or neutral.**

Conclusion

- NHS system is:
  - **High demand & high utilisation**
  - **Growing pressure over time**
  - **Supported by a shared workforce model**
  - **Reliant on traditional (face-to-face) delivery**
  - **Tweets are informative but not consistently engaging or widely shared**

- ⚠️ **Data quality improvements needed** for better planning

Recommendations